

Part Lifecycle Tracker

User Guide

v0 · June 2026

Track every part — from design artifact to certified finished unit.

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1. Introduction

What is the Part Lifecycle Tracker?

The Part Lifecycle Tracker is a full-stack manufacturing application that follows a part across its complete lifecycle — from a customer placing an order, through supplier purchasing and physical receipt, through assembly with enforced build order and dual sign-off, to a certified finished unit.

It is purpose-built around a single insight: **the design of a part and the physical objects built from that design are fundamentally different things** that need to be tracked separately. Most manufacturing tools conflate them. When that distinction is lost, answering questions like *"Did serial HMR-0047 get the Rev C dome or the Rev B one, and who validated it?"* requires digging through spreadsheets. This system makes those questions trivial.

The Central Problem It Solves

Manufacturing operations depend on two kinds of knowledge about a part: what it is *supposed* to be (the design artifact) and what physically exists in the world (the physical object). When those are stored in the same record, you can't trace history — you only know the current state.

- **Without this system:** 'A dome was installed' exists in someone's memory or a spreadsheet cell that got overwritten.
- **With this system:** Every installation, validation, and certification is an immutable timestamped event. You can replay the entire history of any serial number.

Key Capabilities

- Define parts with bills of materials (BOMs) and revision control
- Create purchase orders and receive physical parts — generating serialized instances automatically
- Assemble work orders with enforced build-order dependencies (the system blocks you from installing the dome before the muzzle — no exceptions)
- Enforce a four-eyes validation workflow: installation, independent validation, and QA certification are three separate acts by three different roles
- Record test results against finished units
- Query a single **/context** endpoint that returns a complete snapshot of any part — designed to be readable by both humans and AI agents
- Real passwordless authentication (Stytch) with server-assigned roles — every action is stamped with the authenticated identity

Who Uses It

Five roles cover the full demand-to-fulfillment loop, exercised by six seeded demo personas — Jamie and Riley share the Installer/Validator role. You sign in as a persona (one click on the login screen — see Chapter 3), and the server assigns your role from your identity. Each role unlocks role-appropriate screens and action buttons; everything you do is recorded under your email.

Role	Demo Persona	Primary Responsibility
Salesperson	Sarah Chen	Creates customer orders
Floor Manager	Marcus Webb	Oversees the assembly floor
Installer/Validator	Jamie Torres	Installs parts, and validates parts Riley installed
Installer/Validator	Riley Park	Installs parts, and validates parts Jamie installed
QA Engineer	Dr. Quinn	Certifies that assembly steps functionally work
Site Manager	Alex Reyes	Cross-cutting oversight of all operations

■ **One nuance:** Jamie and Riley hold the same role — **Installer/Validator** (shown as "Installer" in the app header). It is a single permission set: either of them can install, and the four-eyes rule is enforced per *identity* — whoever installed a step cannot validate it. Jamie validates parts Riley installed, and Riley validates parts Jamie installed.

2. Core Concepts

Part Definitions vs. Part Instances

The entire system is built on one foundational split:

Part Definition	Part Instance
The design artifact. One per part number (e.g. THE-HOMER-001). Contains the bill of materials, revision, and status.	A physical object with a serial number (e.g. HMR-0047). Many instances per part number. Has a complete, immutable history.
Has its own lifecycle: DRAFT → RELEASED → OBSOLETE.	Has its own event log: ORDERED → RECEIVED → INSPECTED → IN_ASSEMBLY → INSTALLED → VALIDATED → CERTIFIED.
Locked when RELEASED — revision and BOM cannot be edited.	Never deleted. Events are append-only and never modified.

The Bill of Materials (BOM)

A Part Definition's BOM is a list of child parts required to assemble it. Each BOM item specifies a child part and a quantity. BOM items can have **dependencies** — prerequisites that must be completed before that item can be installed.

■ **Example:** The Homer (THE-HOMER-001) has two BOM items: MUZZLE-001 and DOME-TRANS-001. DOME-TRANS-001 has MUZZLE-001 as a prerequisite — you cannot install the dome until the muzzle is installed, validated, and certified. The system enforces this with a hard block (409 error), not a warning.

Lifecycle Events

Every state change on a Part Instance is recorded as an immutable lifecycle event. Events are never updated or deleted. The current status of an instance is always derived from the most recent event. Every event records:

- **Event type** — what happened (RECEIVED, INSTALLED, VALIDATED, etc.)
- **Actor** — who performed the action, stamped automatically from the authenticated session (there is no "who are you" field to fill in)
- **occurred_at** — when the event happened in the real world (can be backdated for paper-based records)

- **recorded_at** — when the system recorded it (set by the server, never editable — the audit anchor)
- **Notes** — optional free-text annotation

Status	Description
ORDERED	Part is on a supplier purchase order; not yet received
RECEIVED	Part arrived and was logged into the system
INSPECTED	Part passed incoming inspection
IN_ASSEMBLY	Part is being used in an active work order
INSTALLED	Part installed into a work order step by a technician
VALIDATED	Installation validated by a second technician
CERTIFIED	Step certified by QA — part is functionally accepted

The Four-Eyes Principle

Assembly steps require **three separate acts by different people**:

- **Install** — a technician installs the part and records the serial number
- **Validate** — a *different* technician confirms the installation was done correctly. The system rejects the same person validating their own work (409 SAME_ACTOR)
- **Certify** — the QA Engineer confirms the step functionally works

■ **Why this matters:** these are exactly the questions an auditor asks about a part. *When was it installed, and by whom? Who validated that it was installed correctly — and when? Who certified that it works — and when?* Each act records its own actor-and-timestamp signature on the step, so every answer is a lookup, not an interview. Encoding that in the system is safer than trusting a checklist.

The Demand-to-Fulfillment Loop

The full flow from customer to certified unit follows this sequence:

#	Event	Who / What
1	Customer Order created	Salesperson creates order with line items (part + qty)

2	BOM Explosion (async)	System checks stock; reserves available parts or raises supplier PO lines for shortfall
3	Supplier PO received	Warehouse receives physical shipment; instances are created automatically
4	Work Order opened	Floor Manager opens a work order for a specific serial number
5	Assembly steps	Techs install, validate, certify each BOM step in dependency order
6	Work Order completed	All steps certified; work order closed

3. The User Interface

Signing In

Authentication is real (Stytch passwordless). Opening the app takes you to a login screen with two ways in:

- **Explore as a demo user** — six one-click persona buttons. Click one and you are signed in as that seeded user, no email round-trip. This is the path this guide assumes.
- **Email magic link** — enter any email and click **Send magic link**; the emailed link (valid 60 minutes) signs you in. An email that isn't a seeded user still authenticates but gets a **read-only session**: every write control is disabled ("Your session is read-only").

Figure: The login screen: one-click demo personas, or a real email magic link.

Once signed in, the header shows **"Acting as Name (Role)"** next to a **Log out** button. Roles are assigned server-side to each identity — there is nothing to pick. To work as a different persona, log out and choose another demo user. Sessions last about an hour; when one expires the app returns you to the login screen.

■ **A deliberate demo choice:** the one-click demo logins exist so anyone can explore every role without an invitation or an email round-trip. The authentication underneath is real (each click mints a genuine Stytch session), but the demo users themselves are not a production pattern — they would be disabled before a production deployment, leaving magic-link sign-in for real, invited users.

■ ■ **Important:** Your assigned role determines what you can do. Buttons you are not permitted to use stay visible but disabled — hover over the disabled button to see the reason (e.g. "Certify requires the QA Engineer role", "Your role doesn't permit this action").

Navigation

The application has three main tabs accessible from the top navigation bar:

- **Assembly Line** — the assembly floor view: every work order with its status and a per-step tally; click one to open its steps. Primary screen for Installers, Validators, and QA.
- **Sales** — customer orders and their fulfillment status. Live for the Salesperson and Site Manager; every other role sees the tab locked with a tooltip: "Requires the Salesperson or Site Manager role."
- **Parts** — the parts catalog: part definitions, BOMs, and instance counts.

Part Lifecycle Tracker Assembly Line Parts Sales

Acting as Jamie Torres (Installer) Log out

Requires the Salesperson or Site Manager role

Assembly Line

Work orders on the floor. Select one to install, validate, and certify its steps.

8 work orders

SERIAL	PART	STATUS	STEPS
HOMER-CO0001-005	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-004	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-003	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-002	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-001	THE-HOMER-001	OPEN	4 PENDING
HMR-0007	THE-HOMER-001	BLOCKED	4 BLOCKED
HMR-0006	THE-HOMER-001	OPEN	1 PENDING 1 BLOCKED 1 INSTALLED 1 CERTIFIED
HMR-0047	THE-HOMER-001	COMPLETE	4 CERTIFIED

Figure: Signed in as an installer: the Sales tab is locked, with the tooltip explaining why.

</

Figure: The Assembly Line lists every work order with its status and per-step tally.

The Five Screens

Parts List

A searchable table of all Part Definitions with color-coded status badges (DRAFT = gray, RELEASED = green, OBSOLETE = muted red). The 'New Part' button opens a creation form. Clicking any row navigates to Part Detail.


Part Lifecycle Tracker
Assembly Line
Parts
Sales
Acting as Alex Reyes (Site Manager)
Log out

Parts

New Part

9 parts

PART NUMBER	NAME	REVISION	STATUS
BART-HOT-001	Bart's Hot Rod	—	DRAFT
DOVE-TRANS-001	Transparent Dome	A	RELEASED
ENGINE-V8-001	V8 Engine	A	RELEASED
FLANDERS-MOBILE-001	Flanders Family Wagon	A	OBSOLETE
HOMER-CLASSIC-001	The Homer (Classic)	A	OBSOLETE
HORN-LCC-001	La Cucaracha Horn	A	RELEASED
MARGE-FAM-001	Marge's Family Cruiser	—	DRAFT
MUZZLE-001	Muzzle	A	RELEASED
THE-HOMER-001	The Homer	B	RELEASED

Figure: The Parts catalog — searchable, with DRAFT / RELEASED / OBSOLETE badges.

Part Detail

Shows the part number, name, revision, and status with transition controls (**Mark as Released** / **Mark as Obsolete** / **Mark as Draft** buttons, enabled per the state machine). The BOM is displayed as a flat list. On a DRAFT part an **Add BOM item** editor lets you add child parts (with optional prerequisites) and remove them; on a RELEASED part the BOM is locked. A 'View /context' button shows the raw AI-agent snapshot.


Part Lifecycle Tracker
Assembly Line
Parts
Sales

Acting as Dr. Quinn (QA Engineer)
Log out

Parts / THE-HOMER-001

THE-HOMER-001 — The Homer

Revision B · RELEASED

The everyman's car — every feature Homer ever wanted, all at once.

Status

Mark as Draft
Mark as Obsolete

Instances by status

INSPECTED: 1
VALIDATED: 1
INSTALLED: 2
ORDERED: 2
IN_ASSEMBLY: 9
CERTIFIED: 3
RECEIVED: 5

Bill of Materials

The BOM is locked because this part is RELEASED. Only DRAFT parts can be edited.

Component	Name	Qty	Deleted
HORN-LCC-001	La Cucaracha Horn	3	—
DOME-TRANS-001	Transparent Dome	1	—
MUZZLE-001	Muzzle	1	—
ENGINE-V8-001	V8 Engine	1	—

[View /context](#)

Figure: Part Detail for THE-HOMER-001 (RELEASED): status, instances-by-status, BOM, /context.

Work Order

The per-work-order assembly view. Shows all steps with role-aware action buttons (Install on a PENDING step, Validate on an INSTALLED step, Certify on a VALIDATED step). A step blocked by an unmet dependency shows the reason inline (e.g. 'Blocked by Muzzle (must be certified first)') — no hovering or guessing required. The Complete Work Order button is disabled until every step is CERTIFIED.


Part Lifecycle Tracker
Assembly Line
Parts
Sales
Acting as Dr. Quinn (QA Engineer)
Log out

← Assembly Line

Work Order HMR-0006

THE-HOMER-001 · OPEN

Acting as dr.quinn@example.com

MUZZLE-001 Muzzle
Installed by jamie.torres@example.com · Validated by riley.park@example.com

VALIDATED
Certify

DOME-TRANS-001 Transparent Dome
Blocked by Muzzle (must be certified first)

BLOCKED

ENGINE-V8-001 V8 Engine
Installed by jamie.torres@example.com

INSTALLED
Validate

HORN-LCC-001 La Cucaracha Horn

PENDING
Serial #
Install

Complete work order

All steps must be CERTIFIED to complete.

Figure: A work order's steps with role-aware Install / Validate / Certify actions.

Instance Detail

Shows the serial number, current status, the full lifecycle event log (chronological order, both occurred_at and recorded_at visible), and all test records (color-coded PASS/FAIL/INCONCLUSIVE). A **Record event** button appends to the append-only log, and an **Add test result** button records a PASS/FAIL/INCONCLUSIVE test against the unit.


Part Lifecycle Tracker
Assembly Line
Parts
Sales

Acting as Dr. Quinn (QA Engineer)
Log out

INSTANCE

HMR-0016

Current status: CERTIFIED Part: THE-HOMER-001

Record event

Lifecycle Events

- ORDERED

jamie@factory.com

HMR-0016 -> ORDERED

Occurred: May 28, 2026, 04:43 AM Recorded: May 31, 2026, 05:46 PM
- RECEIVED

jamie@factory.com

HMR-0016 -> RECEIVED

Occurred: May 28, 2026, 04:46 AM Recorded: May 31, 2026, 05:46 PM
- INSPECTED

jamie@factory.com

HMR-0016 -> INSPECTED

Occurred: May 28, 2026, 04:47 AM Recorded: May 31, 2026, 05:46 PM
- IN_ASSEMBLY

jamie@factory.com

HMR-0016 -> IN_ASSEMBLY

Occurred: May 28, 2026, 04:48 AM Recorded: May 31, 2026, 05:46 PM
- INSTALLED

jamie@factory.com

HMR-0016 -> INSTALLED

Occurred: May 28, 2026, 04:49 AM Recorded: May 31, 2026, 05:46 PM
- VALIDATED

jamie@factory.com

HMR-0016 -> VALIDATED

Occurred: May 28, 2026, 04:50 AM Recorded: May 31, 2026, 05:46 PM
- CERTIFIED

jamie@factory.com

HMR-0016 -> CERTIFIED

Occurred: May 28, 2026, 04:51 AM Recorded: May 31, 2026, 05:46 PM

Add test result

Test Records

Brake Pressure Test

Conducted by quinn@factory.com

Brake line pressure 12% below spec; rejected for rework.

Occurred: Jun 2, 2026, 11:14 AM Recorded: Jun 2, 2026, 11:14 AM

FAIL

Functional Inspection

Conducted by quinn@factory.com

All systems nominal; horn plays La Cucaracha correctly.

Occurred: Jun 2, 2026, 11:12 AM Recorded: Jun 2, 2026, 11:13 AM

PASS

Figure: Instance Detail: the append-only lifecycle log plus Record-event / Add-test actions.

Sales / Customer Orders

Unlocked for the Salesperson and Site Manager roles (locked for everyone else). Lists all customer orders with status and line counts. 'New Customer Order' lets you create an order with multiple line items. The order detail view shows whether each line is in-stock or pending a supplier PO.



Part Lifecycle Tracker

Assembly Line

Parts

Sales

Acting as Alex Reyes (Site Manager) Log out

Customer Orders

New Customer Order

Lovejoy Parish Vans 8883a99d-e8a6-4b7d-ae1-66278bc8c6e9	OPEN
Springfield Monorail Co 79c794c9-f36d-448a-b17c-d139bb14bb89	COMPLETE
Kwik-E-Mart 93d8d46a-48a4-44d1-8876-65282927e367	OPEN
Elbon Services, Inc 61b6145a-79cd-4233-9683-f093fbbd838e	OPEN
Elbon Services, Inc 65b0622b-68a5-4b7c-b838-4ef2784c967f	OPEN
Quimby Fleet Services 495189a3-1816-46bf-abd9-7478b1f766ae	OPEN
Burns Industries 18c82694-cb49-483b-85ff-7d7fb06fc7ac6	OPEN
Springfield Taxi Co 17344855-44ad-4172-9d41-749274396691	IN_FULFILLMENT
Powell Motors 535a4491-8e69-4b79-b975-8d65c3a8fd86	IN_FULFILLMENT

Figure: The Sales tab — customer orders (Salesperson / Site Manager only).

4. Role: Salesperson

Salesperson

Demo persona: Sarah Chen

As a Salesperson, you create customer orders and track their fulfillment. Your job ends at order creation — the system reserves stock and raises supplier purchase orders automatically, and the floor takes it from there.

Your Primary Screen

The **Sales** tab is your home. It lists all customer orders and their current fulfillment status. You can also see a summary of each order's lines — how many are in stock versus waiting on a supplier.

Creating a Customer Order

Click **New Customer Order** from the Sales tab.

- Enter the customer name
- Add one or more line items: select the part from the dropdown and enter the quantity
- Submit — the order is created immediately

■ ■ **What happens next:** The system automatically checks stock levels in the background. Parts that are in stock get reserved. Parts that are out of stock trigger a supplier purchase order automatically. You don't need to do anything — just check the order detail page to see the fulfillment status.


Part Lifecycle Tracker
Assembly Line
Parts
Sales
Acting as Sarah Chen (Salesperson)
Log out

Customer Orders

[New Customer Order](#)

New Customer Order

Customer name

Line items

Part
THE-HOMER-001 — The Homer
Quantity
20

+ Add line item

[Create order](#)
[Cancel](#)

Springfield Monorail Co 78c794c9-f36d-488a-81fc-d1398b148b89	COMPLETE								
Kwik-E-Mart 93b8d46a-48a4-4d41-8876-65282927c367	OPEN								
Elbon Services, Inc. 61b6145a-79c8-4233-9683-f693f6b8d838e	OPEN								
Elbon Services, Inc. 65b6e228-68d5-4b7c-b838-4ef2794c967f	OPEN								
Quimby Fleet Services 495189a3-1816-46bf-ab09-7478b1f766ae	OPEN								
Burns Industries 18c82694-c849-483b-85f1-7d7fb65fc7ac6	OPEN								
Springfield Taxi Co 17344855-44ad-41f2-9d41-749274396691	IN_FULFILLMENT								
Springfield Taxi Co IN_FULFILLMENT 17344855-44ad-41f2-9d41-749274396691 <table> <thead> <tr> <th>PART</th> <th>NAME</th> <th>QTY</th> <th>FULFILLMENT</th> </tr> </thead> <tbody> <tr> <td>THE-HOMER-001</td> <td>The Homer</td> <td>20</td> <td>Pending supplier PO</td> </tr> </tbody> </table> Fulfillment is a derived best-effort estimate based on linked supplier purchase orders, not an authoritative order field.	PART	NAME	QTY	FULFILLMENT	THE-HOMER-001	The Homer	20	Pending supplier PO	
PART	NAME	QTY	FULFILLMENT						
THE-HOMER-001	The Homer	20	Pending supplier PO						
Powell Motors 535a4491-8e69-4b79-b975-b665c3a8fd86	IN_FULFILLMENT								

Figure: Creating a customer order: a customer name plus one or more line items.

Monitoring Order Fulfillment

Click any order to expand it. Each line shows:

- **In Stock / Reserved** — stock is available and held for this order
- **Needs Ordering** — a supplier PO has been raised; waiting on delivery
- **IN_FULFILLMENT** — assembly is underway for at least one line
- **COMPLETE** — all lines have been assembled and certified

Part Lifecycle Tracker

Assembly Line

Parts

Sales

Acting as Alex Reyes (Site Manager)

Log out

Customer Orders

New Customer Order

Lovejoy Parish Vans

8883a99d-e8a6-4b7d-ae11-b6278bc8c6e9

OPEN

Springfield Monorail Co

79c794c9-f36d-448a-b17c-d139bb14bb89

COMPLETE

Kwik-E-Mart

93d8d46a-48d4-44d1-8876-65282927e367

OPEN

Elbon Services, Inc

61b6145a-79cd-4233-9683-f093fbd8d838e

OPEN

Elbon Services, Inc

65b0622b-68a5-4b7c-b838-4ef2784c967f

OPEN

Quimby Fleet Services

495189a3-1816-46bf-abd9-7478b1f766ae

OPEN

Burns Industries

18c82694-cb49-403b-85ff-7d7b06fc7ac6

OPEN

Springfield Taxi Co

17344855-44ad-4172-9d41-749274396691

IN_FULFILLMENT

Powell Motors

535a4491-0e69-4b79-b975-0d65c3a8fd86

IN_FULFILLMENT

Powell Motors

IN_FULFILLMENT

535a4491-0e69-4b79-b975-0d65c3a8fd86

PART

NAME

QTY

FULFILLMENT

THE-HOMER-001

The Homer

5

In stock

Fulfillment is a derived best-effort estimate based on linked supplier purchase orders, not an authoritative order field.

Figure: An expanded order: per-line fulfillment (Powell Motors — five Homers, in stock).

Order Status Reference

Status	Description
OPEN	Order placed; fulfillment not yet started
IN_FULFILLMENT	Assembly underway for at least one line
COMPLETE	All lines assembled and certified

Tips

- You cannot edit a customer order after creation — create a new order if quantities change
- The Sales tab is only unlocked for the Salesperson and Site Manager roles — sign in as Sarah Chen (or Alex Reyes) to use it
- Orders you create are stamped with your signed-in identity (sarah.chen@example.com) automatically
- You do not need to create supplier purchase orders manually — the system does this automatically when stock is insufficient

5. Role: Floor Manager

Floor Manager

Demo persona: Marcus Webb

As Floor Manager, you oversee the assembly floor. You watch work orders arrive, monitor step-by-step progress, and close work orders once all steps are certified. You do not perform installations yourself — that's for the technicians, and the system enforces it: step actions are disabled for your role.

Your View

You see the full Assembly Line tab: active work orders in progress, the QA queue, and completed work orders. You have read access to all parts, instances, and orders.

Part Lifecycle Tracker

[Assembly Line](#)[Parts](#)[Sales](#)

Acting as Dr. Quinn (QA Engineer) [Log out](#)

Assembly Line

Work orders on the floor. Select one to install, validate, and certify its steps.

8 work orders

SERIAL	PART	STATUS	STEPS
HOMER-CO0001-005	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-004	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-003	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-002	THE-HOMER-001	OPEN	4 PENDING
HOMER-CO0001-001	THE-HOMER-001	OPEN	4 PENDING
HMR-0007	THE-HOMER-001	BLOCKED	4 BLOCKED
HMR-0006	THE-HOMER-001	OPEN	1 PENDING 1 BLOCKED 1 INSTALLED 1 VALIDATED
HMR-0047	THE-HOMER-001	COMPLETE	4 CERTIFIED

Figure: The Assembly Line: every work order with its status and per-step tally.

Opening a Work Order

Each work order is an assembly job for one specific Part Instance (serial number) — one open build per physical serial. When a salesperson creates a customer order, the background BOM explosion reserves stock and raises supplier POs, but it deliberately does *not* open work orders: reserving fungible stock doesn't pick a specific serial, so work-order creation waits until a physical unit is determinable.

Opening one is your call. In v0 it happens through the API (**POST /work-orders** with the part, the serial, and optionally the customer-order line it fulfills — there is no creation form in the UI yet). The system then creates one step per active BOM item and reserves component stock, blocking any step it can't cover. The demo's HOMER-CO0001-001 ... -005 work orders are exactly this shape: one per unit of a customer order, each linked to its order line for traceability.

Monitoring Progress

The work order screen shows all steps and their current status. A step that is **BLOCKED** shows its reason inline — either a dependency that isn't satisfied yet, or insufficient stock. You can see at a glance what the floor is waiting on.

As Floor Manager you read this screen rather than act on it: the Install and Validate buttons are disabled for you ("Your role doesn't permit this action") and Certify requires QA. When something stalls, you dispatch the right person — Jamie or Riley for floor work, Dr. Quinn to certify.

Work Order Status Reference

Status	Description
OPEN	Assembly in progress
BLOCKED	One or more steps blocked by dependency or stock
COMPLETE	All steps certified; work order closed

Completing a Work Order

The **Complete Work Order** button becomes active only when every step is in CERTIFIED status. Clicking it closes the work order and finalizes the instance's lifecycle record.

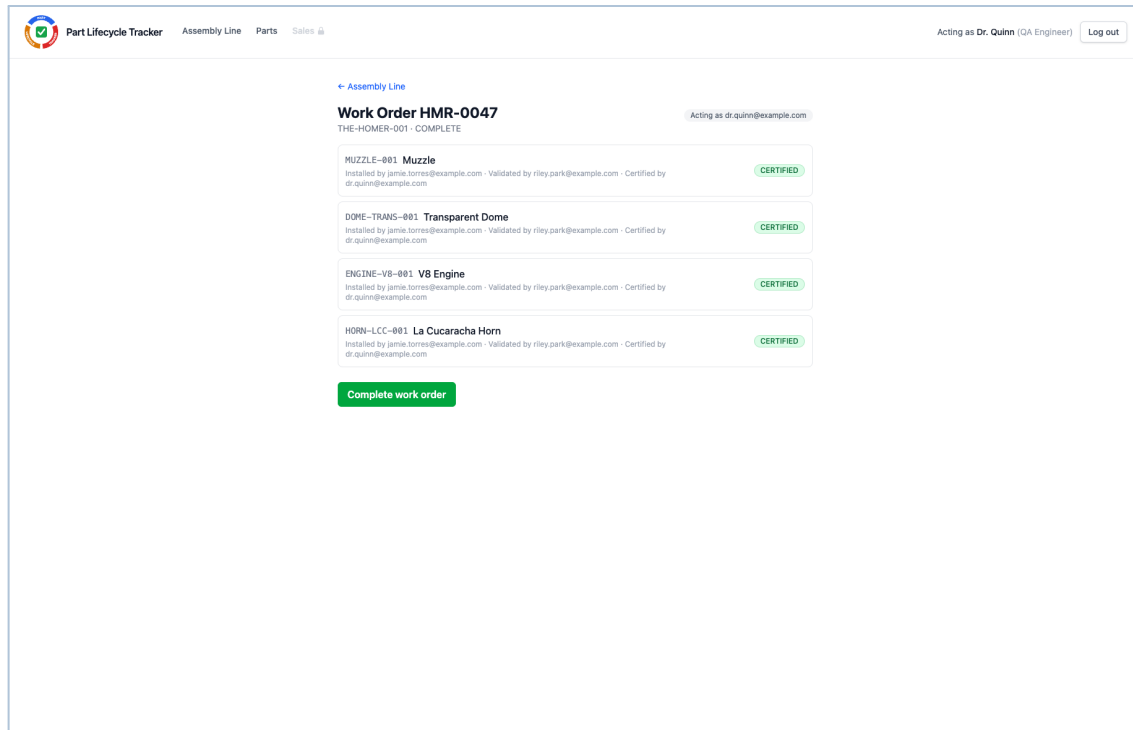


Figure: A completed work order: all steps CERTIFIED, each with its full install / validate / certify history.

Tips

- You cannot complete a work order with any non-certified steps — this is enforced server-side
- Blocked steps must be resolved before assembly can proceed; check the inline reason
- You can view any Part Instance's full event log from the Instance Detail screen

6. Role: Installer/Validator — Installing

Installer/Validator

Demo persona: Jamie Torres

Jamie and Riley share the Installer/Validator role; this chapter follows Jamie through the installing half of the job. You physically install parts into assembly steps and record the serial number of each installed part, following the BOM dependency order — the system will not let you skip ahead. You can also validate, but only parts Riley installed (four-eyes).

Your Primary Screen

The **Assembly Line** tab shows all open work orders. Click any work order to see its steps. Your action is the **Install** button on any PENDING step.

Installing a Part

When a step is PENDING and no dependency is blocking it, you'll see an **Install** button next to it.

- Type the serial number of the physical part you are installing into the step's **Serial #** field — it must be an existing instance, or the server rejects it
- Click **Install** — the step moves to INSTALLED, stamped "Installed by jamie.torres@example.com" from your session

■ Install and Validate are enabled only for the two **Installer/Validator** accounts (Jamie and Riley). Anyone else sees the buttons disabled: "Your role doesn't permit this action."

■ **Blocked steps:** If a step shows 'Blocked by [Part Name] (must be certified first)', a prerequisite step must be completed first — that step offers no Install field until the prerequisite is certified. Complete the prerequisite (install → validate → certify it) before this step opens up.

Part Lifecycle Tracker
Assembly Line
Parts
Sales

Acting as Jamie Torres (installer)
Log out

← Assembly Line

Work Order HOMER-CO0001-001
Acting as jamie.torres@example.com

THE-HOMER-001 - OPEN

MUZZLE-001 Muzzle
Installed by jamie.torres@example.com
INSTALLED
Validate

ENGINE-V8-001 V8 Engine
PENDING
Serial #
Install

HORN-LCC-001 La Cucaracha Horn
Installed by jamie.torres@example.com
INSTALLED
Validate

DOHE-TRANS-001 Transparent Dome
Blocked by Muzzle (must be certified first)
BLOCKED
Install

Complete work order
All steps must be CERTIFIED to complete.

Figure: An installer's view: the steps you installed show a disabled Validate button (four-eyes), the PENDING engine offers a serial field, and the dome is held by its muzzle prerequisite.

Part Lifecycle Tracker
Assembly Line
Parts
Sales

Acting as Jamie Torres (installer)
Log out

← Assembly Line

Work Order HOMER-CO0001-001
Acting as jamie.torres@example.com

THE-HOMER-001 - OPEN

MUZZLE-001 Muzzle
Installed by jamie.torres@example.com
INSTALLED
Validate

ENGINE-V8-001 V8 Engine
PENDING
Serial #
Install

HORN-LCC-001 La Cucaracha Horn
PENDING
Serial #
Install

DOHE-TRANS-001 Transparent Dome
Blocked by Muzzle (must be certified first)
BLOCKED
Install

Complete work order
All steps must be CERTIFIED to complete.

Figure: After installing: the step flips to INSTALLED, stamped with your email from the session.

Dependency Order

The BOM defines which steps must be completed before others. The system strictly enforces this order — attempting to install a dependent step before its prerequisite is complete will be rejected. This is not a warning; it is a hard block.

■ **Example:** On The Homer assembly, you must install and certify the Muzzle before you can install the Dome. If you try to install the Dome first, the system returns an error naming the blocking part.

After Installation

Once you submit an installation, the step is **INSTALLED**. Its **Validate** button is shown but *disabled for you* ('4-eyes: you installed this step and cannot validate it') — validation is reserved for a different technician. Your work on this step is done. Move to the next **PENDING** step.

Step Status Reference

Status	Description
PENDING	Step not yet started
BLOCKED	Dependency unmet or stock unavailable
INSTALLED	You installed this step; awaiting validation
VALIDATED	Validated by a second tech; awaiting QA
CERTIFIED	QA certified; step complete

7. Role: Installer/Validator — Validating

Installer/Validator

Demo persona: Riley Park

Jamie and Riley share the Installer/Validator role; this chapter follows Riley through the validating half of the job — the independent second check on every installation. You can install parts too, but the four-eyes rule is enforced per identity: Validate is enabled only on steps the *other* technician installed (you validate Jamie's installs, Jamie validates yours), and disabled on your own.

Your Primary Screen

The **Assembly Line** tab shows all open work orders. You are looking for steps in **INSTALLED** status that you did not install yourself. Those steps show a **Validate** button.

Validating an Installation

When a step is INSTALLED and someone else performed the installation, you'll see a **Validate** button.

- Physically inspect the installation — confirm it matches the work order requirements
- Click **Validate**
- Confirm — the step moves to VALIDATED status

■ **Four-eyes rule:** You cannot validate a step you installed yourself. On a step you installed, the Validate button is disabled with the note '4-eyes: you installed this step and cannot validate it'; the rule is also enforced server-side, so it cannot be bypassed.

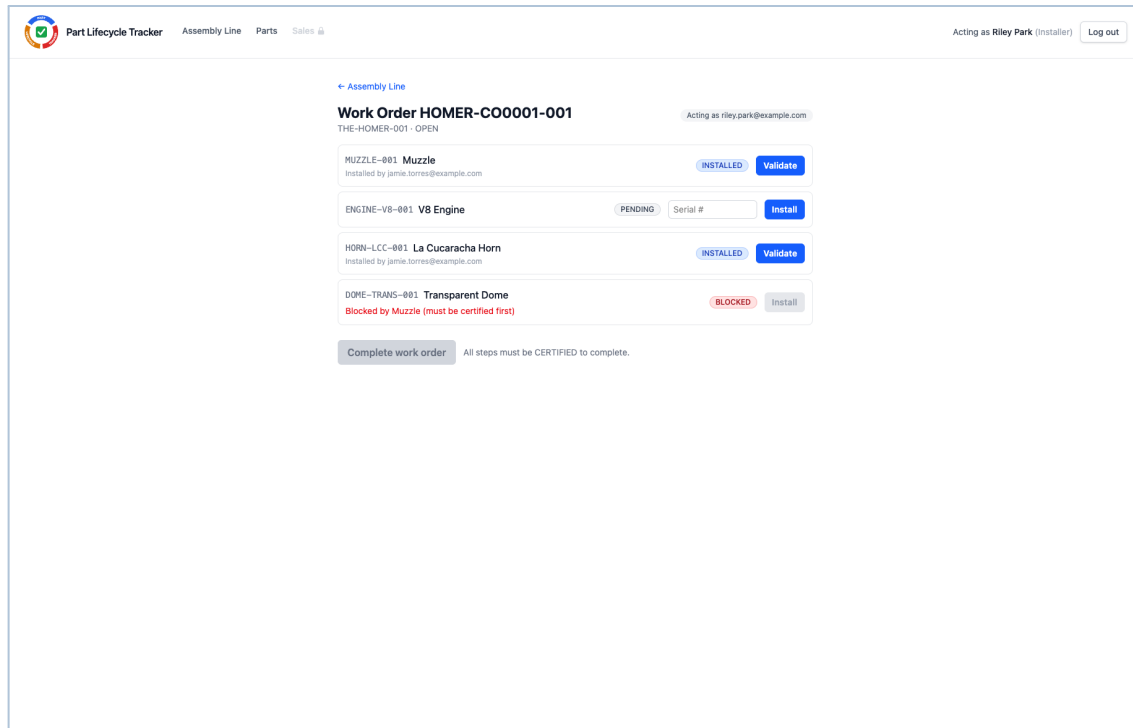


Figure: A validator's view: Validate is enabled for steps a different person installed.

What You Are Confirming

Validation answers the question: *"Was this part installed correctly?"* — not just whether it was installed at all. You are a second set of eyes confirming the work meets quality standards before it goes to QA.

After Validation

Once validated, the step moves to **VALIDATED** and becomes visible to the QA Engineer for certification. Your work on this step is complete.

Tips

- Both Jamie and Riley can install **PENDING** steps — the **Validate** button is simply disabled for whoever installed it
- If every **INSTALLED** step has a disabled **Validate** button, you installed them all — a teammate must validate, or you swap who installs
- You can view the full event log of any instance to see exactly who installed what and when

8. Role: QA Engineer

QA Engineer

Demo persona: Dr. Quinn

As QA Engineer, you are the final gate in the assembly process. Only you can certify a validated step. Your certification means the step functionally works — not just that it was installed and visually confirmed.

Your Primary Screen

The **Assembly Line** tab shows all open work orders. You are looking for steps in **VALIDATED** status. Only the QA Engineer role sees the **Certify** button on validated steps.

Certifying a Step

When a step is **VALIDATED**, the QA Engineer sees a **Certify** button.

- Review the installation and validation history on the Instance Detail screen
- Perform any functional tests required
- Record test results (see below)
- Click **Certify** on the step
- Confirm — the step moves to **CERTIFIED** status

■ **What you are confirming:** Certification answers the question 'Does this assembly step *functionally work?*' — distinct from whether it was installed (the installing technician) or installed correctly (the validating technician). All three questions require different expertise.

Part Lifecycle Tracker Assembly Line Parts Sales

Acting as Dr. Quinn (QA Engineer) Log out

← Assembly Line

Work Order HOMER-CO0001-001 Acting as dr.quinn@example.com

THE-HOMER-001 - OPEN

MUZZLE-001 **Muzzle** CERTIFIED
 Installed by jamie.torres@example.com - Validated by riley.park@example.com - Certified by dr.quinn@example.com

ENGINE-V8-001 **V8 Engine** PENDING Serial # Install

HORN-LCC-001 **La Cucaracha Horn** VALIDATED Certify
 Installed by jamie.torres@example.com - Validated by riley.park@example.com

DOME-TRANS-001 **Transparent Dome** PENDING Serial # Install

Complete work order All steps must be CERTIFIED to complete.

Figure: After you certify the muzzle: the step carries the full install (Jamie) / validate (Riley) / certify (Quinn) chain, and the dome step's dependency hold clears.

Recording Test Results

From the Instance Detail screen, you can record test results against any Part Instance:

- Navigate to the Instance Detail screen for the serial being tested
- Click **Add test result**
- Enter: test type, result (PASS / FAIL / INCONCLUSIVE), notes, and when the test occurred — the conducted-by line is stamped from your session ("Conducted by dr.quinn@example.com — recorded from your session")

■ Adding test results is QA-only — the server rejects any other role. Recording plain lifecycle events, by contrast, is open to every signed-in role.

Test type

Result

Occurred at

Brake Pressure Retest

PASS

06/03/2026, 03:23 PM

Conducted by dr.quinn@example.com — recorded from your session.

Notes (optional)

Brake line re-plumbed; pressure now 3% above minimum spec.

Add test result

Cancel

Figure: Recording a test result: type, PASS/FAIL/INCONCLUSIVE, notes — conducted-by comes from your session.

Test Records	Add test result
Brake Pressure Retest Conducted by dr.quinn@example.com Brake line re-plumbed; pressure now 3% above minimum spec. Occurred: Jun 3, 2026, 03:23 PM Recorded: Jun 3, 2026, 03:23 PM PASS	
Brake Pressure Test Conducted by quinn@factory.com Brake line pressure 12% below spec; rejected for rework. Occurred: Jun 2, 2026, 11:14 AM Recorded: Jun 2, 2026, 11:14 AM FAIL	
Functional Inspection Conducted by quinn@factory.com All systems nominal; horn plays La Cucaracha correctly. Occurred: Jun 2, 2026, 11:12 AM Recorded: Jun 2, 2026, 11:13 AM PASS	

Figure: Test records on a unit — a fail-and-retest story, each record badged and timestamped with who conducted it.

Status	Description
PASS	Test passed; unit meets acceptance criteria
FAIL	Test failed; unit does not meet criteria
INCONCLUSIVE	Test result indeterminate; re-test required

Completing the Work Order

Once **all steps** in a work order are CERTIFIED, the Floor Manager can complete the work order. If any step is not CERTIFIED, the Complete button remains disabled. Your certifications are the final gate.

Tips

- Only the QA Engineer role can certify steps — for everyone else the button is disabled with "Certify requires the QA Engineer role"
- You no longer install or validate: those buttons are disabled for QA ("Your role doesn't permit this action") — certification is your lane
- Test records are stored separately from lifecycle events and are never deleted
- INCONCLUSIVE test results do not block certification — but they are visible in the Instance Detail history

9. Role: Site Manager

Site Manager

Demo persona: Alex Reyes

As Site Manager, you can see everything in the system — including the Sales tab, which is locked for the floor roles. Your job is oversight: checking compliance, tracing serial numbers, monitoring customer order fulfillment, and using the `/context` endpoint for ad-hoc queries.

What You Can See

Site Manager is a read-mostly oversight role with access to everything:

- **Assembly Line tab** — all work orders, all steps, full step history
- **Sales tab** — all customer orders and fulfillment status (unlocked only for you and the Salesperson)
- **Parts tab** — full parts catalog with BOMs and instance counts
- **Instance Detail** — complete event log and test record history for any serial

You can also act where oversight needs it: create customer orders, edit a DRAFT part's BOM, change part statuses, and record lifecycle events. What you can't do is floor work — install/validate are Installer-only, and certify and test results belong to QA.

Using the `/context` Endpoint

Every Part Definition has a **View `/context` Response** button on its detail page. This returns a structured JSON snapshot of the part — its current status, BOM, all instances and their current states, and recent events. The snapshot is designed to be readable by both humans and AI agents.

■ **AI-ready:** The `/context` response includes a pre-rendered summary field. You can paste the entire response into an LLM and ask questions like 'Which instances are in assembly?' or 'Who certified serial HMR-0047?' — and get accurate answers from a single API call.

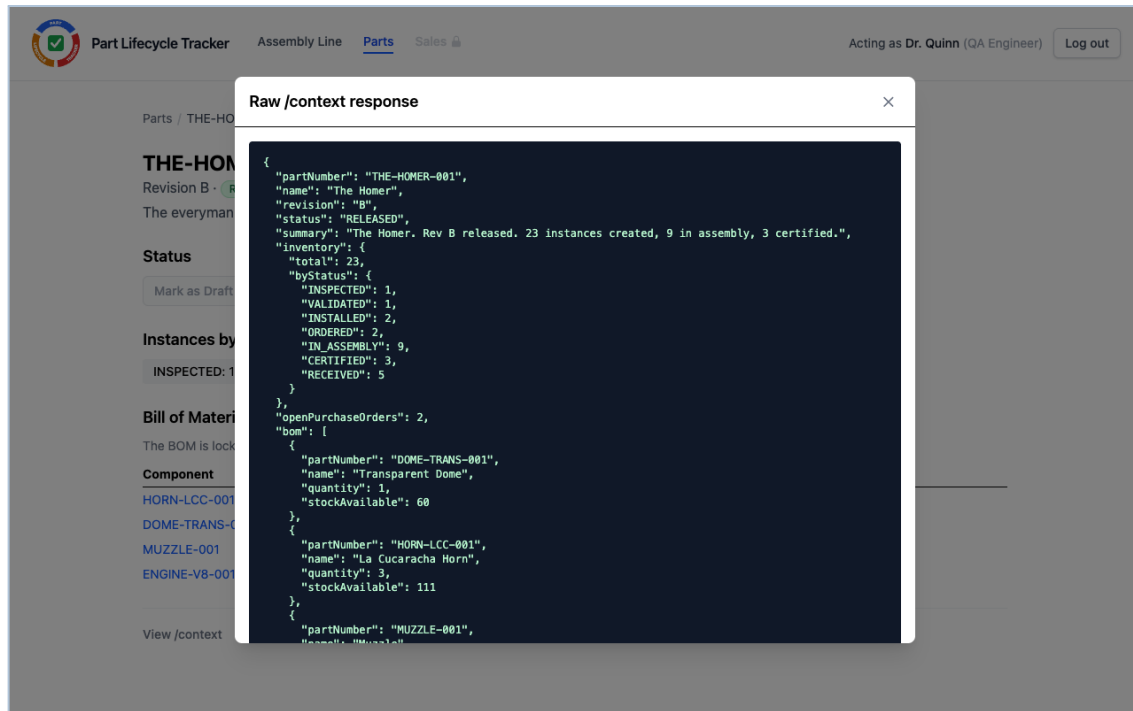


Figure: The /context snapshot — deterministic JSON (templated summary, inventory, BOM with stock, recent events) for humans and AI agents alike.

Oversight Scenarios

Checking build order compliance

Open any work order on the Assembly Line tab. Blocked steps show their reason inline. The Instance Detail screen shows the full event log with actor names and timestamps — you can see exactly who did what and when.

Tracing a specific serial number

From the Parts tab, click a part definition and scroll to the instances section. Click any serial number to open Instance Detail. The event log is the complete audit trail — every RECEIVED, INSTALLED, VALIDATED, and CERTIFIED event is there, with actor and timestamp.

Monitoring customer order fulfillment

The Sales tab shows all customer orders. Click an order to see line-by-line fulfillment status — whether each line is in stock, pending a supplier PO, or in assembly.

10. Appendix: Status Reference

Part Definition Statuses

Status	Description
DRAFT	New part; BOM and revision are editable
RELEASED	BOM and revision are locked; instances can be created
OBSOLETE	Part is retired; no new instances; existing instances remain

Part Instance Event Sequence

Events on a Part Instance flow in this order. Every event is immutable.

Status	Description
ORDERED	On a supplier purchase order; not yet received
RECEIVED	Part received from supplier
INSPECTED	Incoming inspection passed
IN_ASSEMBLY	Part is in an active work order
INSTALLED	Installed into a work order step
VALIDATED	Installation validated by a second technician
CERTIFIED	QA certified; assembly complete

Work Order Step Statuses

Status	Description
PENDING	Not yet started
BLOCKED	Dependency unmet or stock unavailable; shows reason inline

INSTALLED	Installed; awaiting validation
VALIDATED	Validated; awaiting QA certification
CERTIFIED	QA certified; step complete

Test Record Results

Status	Description
PASS	Unit meets acceptance criteria
FAIL	Unit does not meet acceptance criteria
INCONCLUSIVE	Result indeterminate; re-test required

Common Error Codes

Code	HTTP	Meaning
DEPENDENCY_NOT_MET	409	A BOM prerequisite step is not yet certified. The error names the blocking part.
SAME_ACTOR	409	The same person who installed a step attempted to validate it. Four-eyes rule enforced.
INVALID_TRANSITION	422	The requested status transition is not allowed (e.g. RELEASED → DRAFT).
REVISION_LOCKED	422	Attempted to edit the revision field on a non-DRAFT part.
UNAUTHENTICATED	401	No valid session — the Stytlch session JWT is missing, expired, or revoked. The app returns you to the login screen.
READ_ONLY	403	Authenticated, but the identity is not a seeded user, so it has no assigned role. Reads work; all writes are refused.
FORBIDDEN	403	Your assigned role does not permit this action (e.g. a non-QA account attempting to certify or record a test result).

Glossary

Part Definition

The design artifact for a part — one per part number. Contains BOM, revision, and status.

Part Instance

A physical serialized object — many per part number. Has an immutable event log.

BOM

Bill of Materials — the list of child parts required to assemble a parent part.

BOM Dependency

A prerequisite relationship between two BOM items: item B cannot be installed until item A is certified.

Lifecycle Event

An immutable record of a state change on a Part Instance. Never updated or deleted.

occurred_at

When an event happened in the real world. Can be backdated for paper records.

recorded_at

When the system recorded the event. Set by the server; cannot be modified.

Four-Eyes Principle

The rule that install, validate, and certify must be performed by different people — enforced per authenticated identity.

Assigned Role

The role attached server-side to an authenticated identity (Salesperson, Floor Manager, Installer/Validator, QA Engineer, Site Manager — five roles, six personas). Never chosen in the client.

Demo Login

A one-click button on the login screen that mints a real Stytc session for one of the six seeded personas.

Read-only Session

An authenticated session whose email is not a seeded user: it has no role, can browse everything, and can write nothing.

Work Order

An assembly job for a specific Part Instance, walking through each BOM step.

Work Order Step

One BOM item in a work order. Progresses PENDING → INSTALLED → VALIDATED → CERTIFIED.

BOM Explosion

The async process of reserving stock or raising supplier POs when a customer order is created.

/context endpoint

A single API endpoint that returns a complete snapshot of a part — designed for AI agent consumption.